The documentation and process conversion measures necessary to comply with this revision shall be completed by 15 March 1998.

INCH POUND

MIL-PRF-19500/453C
15 December 1997
SUPERSEDING
MIL-S-19500/453B
27 May 1994

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, HIGH-FREQUENCY
TYPE 2N5109, JAN, JANTX, AND JANTXV

This specification is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for NPN silicon, VHF-UHF amplifier transistors. Three levels of product assurance are provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See 3.3 (T0-39).

1.3 Maximum ratings.

P_T 1/ 2/ T_A = +25° C	V_{CBO}	V_{CEO}	I_C	V_{EBO}	T_{STG}	T_J
W	V dc	V dc	A dc	V dc	°C	°C
1	40	20	0.4	3.0	-65 to +200	+200

1/ Derate at 5.71 mW/° C above T_A > +25° C.
2/ P_T = 2.9 W at T_C = +25° C, derate at 16.6 mW/° C above T_C > +25° C.

1.4 Primary electrical characteristics (common to all types).

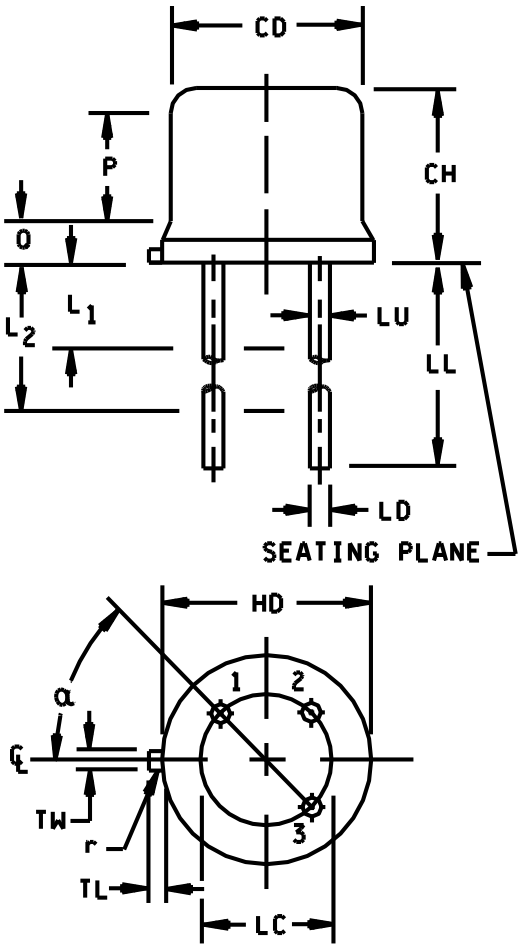
Limits	h_{FE}	$V_{CE(SAT)}$	C_{obo}	$\|h_{FE}\|$	Power gain
	V_{CE} = 15 V dc I_C = 50 mA dc	I_C = 100 mA dc I_B = 10 mA dc	I_E = 0 100 kHz ≤ f ≤ 1 MHz V_{CB} = 28 V dc	V_{CE} = 15 V dc I_C = 50 mA f = 200 MHz	I_C = 50 mA dc f = 200 MHz Pin = -10 dBm V_{CE} = 15 V dc
		V dc	pF		dB
Min	40			6.0	11.0
Max	150	0.5	3.5	11.0	

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad St., Columbus, OH 43216-5000, by using the addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
LC	.190	.210	4.83	5.33	
CH	.240	.260	6.10	6.60	
LD	.016	.021	0.406	0.533	4
LU	.016	.019	0.406	0.483	4
HD	.335	.370	8.51	9.40	
CD	.305	.335	7.75	8.51	
TW	.028	.034	0.711	0.864	
TL	.029	.045	0.737	1.14	5
LL	.500	.750	12.70	19.05	4
L_1		.050		1.27	4
L_2	.250		6.35		4
P	.100		2.54		3
Q		.050		1.27	6
α	45° T. P.				

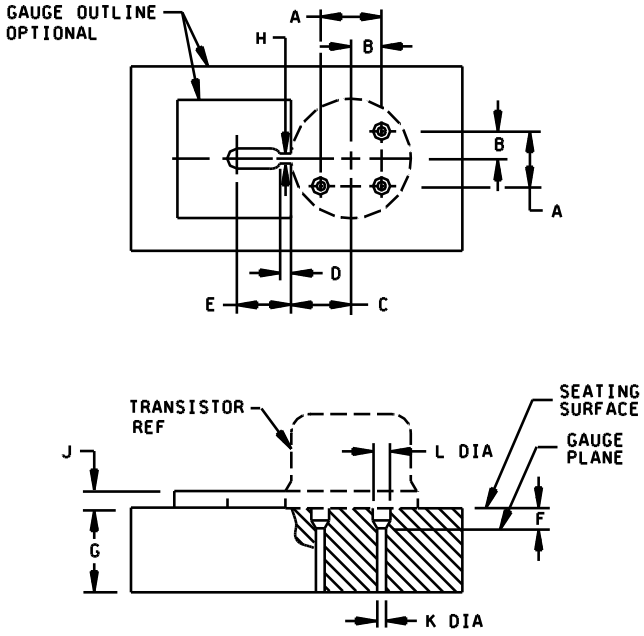


NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. This zone is controlled for automatic handling. The variation in actual diameter within this zone shall not exceed .010 inch (0.254 mm).
4. (Three leads) LU applies between l_1 and l_2. LD applies between l_2 and .5 inch (12.70 mm) from seating plane. Diameter is uncontrolled in l_1 and beyond .5 inch (12.70 mm) from seating plane.
5. Measured from maximum diameter of the actual device.
6. Details of outline in this zone optional.

FIGURE 1. Physical dimensions of transistor type 2N5109.

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Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.1409	.1419	3.579	3.604
B	.0702	.0712	1.783	1.809
C	.182	.199	4.62	5.05
D	.009	.011	0.23	0.28
E	.125 Nom		3.18 Nom	
F	.054	.055	1.37	1.40
G	.372	.378	9.45	9.60
H	.0350	.0355	0.889	0.902
J	.150 Nom		3.81 Nom	
K	.0325	.0335	0.826	0.851
L	.0595	.0605	1.511	1.537



NOTES:
1. Metric equivalents are given for general information only.
2. Dimensions are in inches.
3. The following gauging procedures shall be used: The use of a pin straightener prior to insertion in the gauge is permissible. The device being measured shall be inserted until its seating plane is .125 inch (3.18 mm) +.010 inch (0.254 mm) from the seating surface of the gauge. A spacer may be used to obtain the .125 inch (3.18 mm) distance from the gauge seat prior to force application. A force of 8 ±.5 oz. shall then be applied parallel and symmetrical to the device's cylindrical axis. When examined visually after the force application (the force need not be removed) the seating plane of the device shall be seated against the gauge.
4. The location of the tab locator, within the limits of dimension C, will be determined by the tab and flange dimension of the device being checked.

FIGURE 2. Gauge for lead and tab location.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Defense Printing Service Detachment Office, Bldg. 4D (Customer Service), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated specification. The individual item requirements shall be in accordance with MIL-PRF-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.3 Interface requirements and physical dimensions. The Interface requirements and physical dimensions shall be as specified in MIL-PRF-19500, and figure 1 (T0-39) herein.

3.3.1 Lead finish. Lead finish shall be solderable as defined in MIL-PRF-19500.

3.4 Marking. Marking shall be in accordance with MIL-PRF-19500. At the option of the manufacturer, the country of origin marking and the manufacturers identification may be omitted from the body of the transistor, but shall be retained on the initial container.

3.5 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.2).

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in paragraph 1.3, 1.4, and table I.

3.7 Electrical test requirements. The electrical test requirements shall be the subgroups specified in paragraphs 4.4.2 and 4.4.3.

4. VERIFICATION

4.1 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3)

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500, and as specified herein.

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4.3 Screening (JANTX and JANTXV levels only). Screening shall be in accordance with MIL-PRF-19500 (table IV), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement JANTX and JANTXV levels
3c	Thermal impedance (see 4.3.2)
7	hermetic seal (optional) 1/
9	Not applicable
10	48 hours minimum
11	I_{CEO1} and h_{FE1}
12	See 4.3.1 80 hours minimum
13	ΔI_{CEO1} = 100% of initial value or 2 µA dc, whichever is greater. Δh_{FE1} = 20% change from initial reading; Subgroup 2 of table I herein.

1/ Hermetic seal test shall be performed in either screen 7 or screen 14.

4.3.1 Power burn-in. Power burn-in conditions are as follows:
T_A = Room ambient as defined in the general requirements of MIL-STD-750, section 4.5;
V_{CB} = 15 V dc; P_T = 1.0 W.

4.3.2 Thermal impedance ($Z_{\Theta JX}$ measurements). The $Z_{\Theta JX}$ measurements shall be performed in accordance with MIL-STD-750, Method 3131.

a. I_M measurement current - - - - - - - - - - - 5 mA.
b. I_H forward heating current - - - - - - - - 200 mA (min).
c. t_H heating time - - - - - - - - - - - - - 25 - 30 ms.
d. t_{md} measurement delay time - - - - - - - - 60 µs max.
e. V_{CE} collector-emitter voltage - - - - - - - 10 V dc minimum

The maximum limit for $Z_{\Theta JX}$ under these test conditions are $Z_{\Theta JX}$ (max) = 72° C/W.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein. If alternate screening is being performed per E.5.3.1d of MIL-PRF-19500, a sample of screened devices shall be submitted to and pass the requirements of group A1 and A2 inspection only (table VIb, group B, subgroup 1 is not required to be performed again if group B has already been satisfied per 4.4.2).

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in 4.4.2.1 for JAN, JANTX, and JANTXV group B testing. Electrical measurements (end-points) and delta requirements shall be in accordance with the steps of table II herein as specified in the notes for table II.

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4.4.2.1 Group B inspection, (JAN, JANTX, and JANTXV). 1/

Step	Method	Condition
1	1039	HTRB, test condition A, T_A = 150° C, V_{CB} = 32 V dc, 340 hours, n = 22 devices, c = 0.
2	1039	Steady-state life: Test condition B, 1,000 hours, V_{CB} = 15 V dc, P_T = 1 W; at T_A = room ambient per MIL-STD-750. No heat sink or forced-air cooling on the devices shall be permitted. n = 45 devices, Acc = 0
	OR	
	1037	Intermittent operation life: V_{CB} = 15 V dc , 6,000 cycles, $\Delta T_J \geq +100°$ C; forced air cooling allowed on cooling cycle only. n = 45 devices, Acc = 0.
3	2037	Bond strength, n = 11 wires, c = 0.
4	1032	High-Temperature life (non-operating), T_A = +200° C n = 22, c = 0

1/ Separate samples may be used for each step. For rules on resubmission for failed steps, see MIL-PRF-19500 rules on resubmission of failed subgroups.

4.4.2.2 Group B sample selection. Samples selected from group B inspection shall meet all of the following requirements:

a. Samples shall be selected from each wafer lot from a die fabrication line that the manufacturer chooses to "certify" for the die design used in products manufactured to this specification.

b. Must be chosen from an inspection lot that has been submitted to and passed group A conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for life test (group B for JAN, JANTX, and JANTXV) may be pulled prior to the application of final lead finish.

4.4.2.3 Product acceptable for delivery. The product shall be acceptable for delivery only after the successful completion of all group B testing and shall be comprised of die meeting the following requirements:

a. Manufactured on the same certified die fabrication line as the sample selected for 4.4.2.3.

b. The die lot was manufactured (using start or completion dates, as the manufacturer selects to consistently use) within three months following the wafer lot used for sample selection in paragraph 4.4.2.3. Group B shall not be required more than once for any single wafer lot.

4.4.3 Group C inspection, JAN, JANTX, and JANTXV. Group C inspection (JAN, JANTX, and JANTXV, package related tests) shall be as follows and shall include those package or case-related tests which are performed periodically. Group C test on each package family shall be performed on devices from each six months production of devices (based on inspection lot identification codes) for each assembly location and die attach method. Electrical measurements (end points) and delta requirements shall be in accordance with the steps of table II herein as specified in the notes for table II. The sample size for each of these subgroups is: 22 devices, c = 0. Separate samples may be used for each step. For rules on resubmission for failed steps, see MIL-PRF-19500 rules on resubmission of failed subgroups.

Step	Method	Condition
1	2066	
2	1051	Test condition C.
3	2036	Test condition E.

4.4.3.1 Group C sample selection. Samples for subgroups in group C shall be chosen at random from any inspection lot containing the intended package type and lead finish procured to the same specification which is submitted to and passes group A tests for conformance inspection. Testing of a subgroup using a single device type enclosed in the intended package type shall be considered as complying with the requirements for that subgroup.

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4.4.3.4 Product acceptable for delivery. The product which may be accepted for delivery based a successful completion of group C testing shall be product of the particular package type with lot inspection identification codes of the 52 successive weeks beginning with the inspection lot identification code of the successful group C sample (i.e. the inspection lot identification codes accepted for delivery based on successful group C testing will include those sealed during the approximate 10 weeks during which the sample was being screened and tested plus those sealed in the 42 weeks following successful completion of group C testing making a total of 52 successive calendar weeks.) A manufacturer shall not accept product for delivery until after successful completion of the group C testing on product of the applicable package type with an inspection lot identification code less than 51 weeks prior to the inspection identification code of the product being considered for acceptance.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Solderability 1/ 2/	2026	15 leads, c = 0				
Resistance to solvents 2/ 3/	1022	15 devices, c = 0				
Subgroup 2						
Breakdown voltage, collector to base	3001	Bias condition D; I_C = 100 µA dc	$V_{(BR)CBO}$	40		V dc
Breakdown voltage, collector to emitter	3011	Bias condition D; I_C = 5 mA dc	$V_{(BR)CEO}$	20		V dc
Breakdown voltage, emitter to base	3026	Bias condition D; I_C = 100 µA dc	$V_{(BR)EBO}$	3.0		V dc
Breakdown voltage, collector to emitter	3011	Bias condition B; I_C = 5 mA dc; R_2 = 10Ω	$V_{(BR)CER}$	40		V dc
Collector to emitter cutoff current	3041	Bias condition D; V_{CE} = 15 V dc	I_{CE01}		20	µA dc
Forward-current transfer ratio	3076	V_{CE} = 15 V dc; I_C = 50 mA dc; Pulsed (see 4.5.1)	h_{FE1}	40	150	
Collector to emitter saturated voltage	3071	I_C = 100 mA dc; I_B = 10 mA dc; Pulsed (see 4.5.1)	$V_{CE(sat)1}$		0.5	V dc
Subgroup 3						
High temperature operation:		T_C = +175° C				
Collector to emitter cutoff current	3041	Bias condition D; V_{CE} = 15 V dc	I_{CE02}		5.0	mA dc
Low temperature operation:		T_A = -55° C				
Forward-current transfer ratio	3076	V_{CE} = 5.0 V dc; I_C = 50 mA dc; Pulsed (see 4.5.1)	h_{FE2}	15		
Subgroup 4						
Open circuit output capacitance	3236	V_{CB} = 28 V dc; I_E = 0; 100 kHz ≤ f ≤ 1 MHz	C_{obo}		3.5	pF

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 4 - Continued.						
Magnitude of common-emitter small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 15 V dc; f = 200 MHz	$\lvert h_{fe} \rvert$			
		I_C = 25 mA dc;		5.0	10	
		I_C = 50 mA dc;		6.0	11.0	
		I_C = 100 mA dc		5.0	10.5	
Power gain (narrow band) current		V_{CC} = 15 V dc; I_C = 50 mA dc; f = 200 MHz; Pin = -10 dB; (See figure 3)	G_{PE}	11		dB
Cross modulation		V_{CC} = 15 V dc; I_C = 50 mA dc; 54 dBm V output; (See figure 4)	cm		-57	dB
Noise figure		V_{CC} = 15 V dc; I_C = 10 mA dc; f = 200 MHz; (See figure 3)	NF		3.5	dB
Voltage gain (wideband)		V_{CC} = 15 V dc; I_C = 50 mA dc; f = 50 to 216 MHz; (See figure 4)	G_{VE}	11		dB
Subgroups 5, 6, and 7						
Not applicable						

1/ For sampling plan, see MIL-PRF-19500.
2/ Separate samples may be used.
3/ Not required for laser marked devices.

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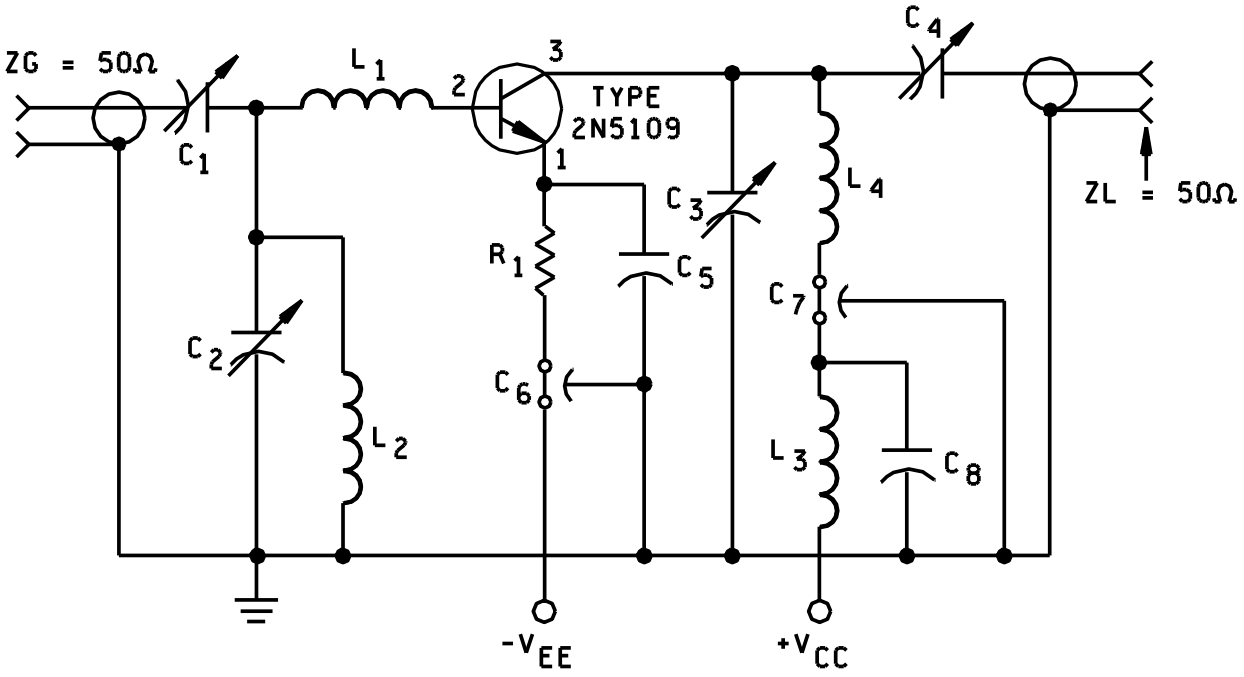
TABLE II. Groups B and C electrical end-point inspection measurements. 1/ 2/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Collector to emitter cutoff current	3041	Bias condition D; V_{CE} = 15 V dc	I_{CE01}		20.0	µA dc
2.	Collector to emitter cutoff current	3041	Bias condition D; V_{CE} = 15 V dc	I_{CE02}		40.0	µA dc
3.	Breakdown voltage, emitter to base	3026	Bias condition D; I_E = 100 µA dc	$V_{(BR)EBO}$	3.0		

1/ The electrical measurements for group B, 4.4.2.1 herein(JAN, JANTX, and JANTXV,) are as follows:
a. Steps 1, 2, and 4, of 4.4.2.1, see table II herein, all steps.

2/ The electrical measurements for group C, 4.4.3 herein (JAN, JANTX, and JANTXV) are as follows:
a. Step 2 of 4.4.3.2, see table II herein, all steps.

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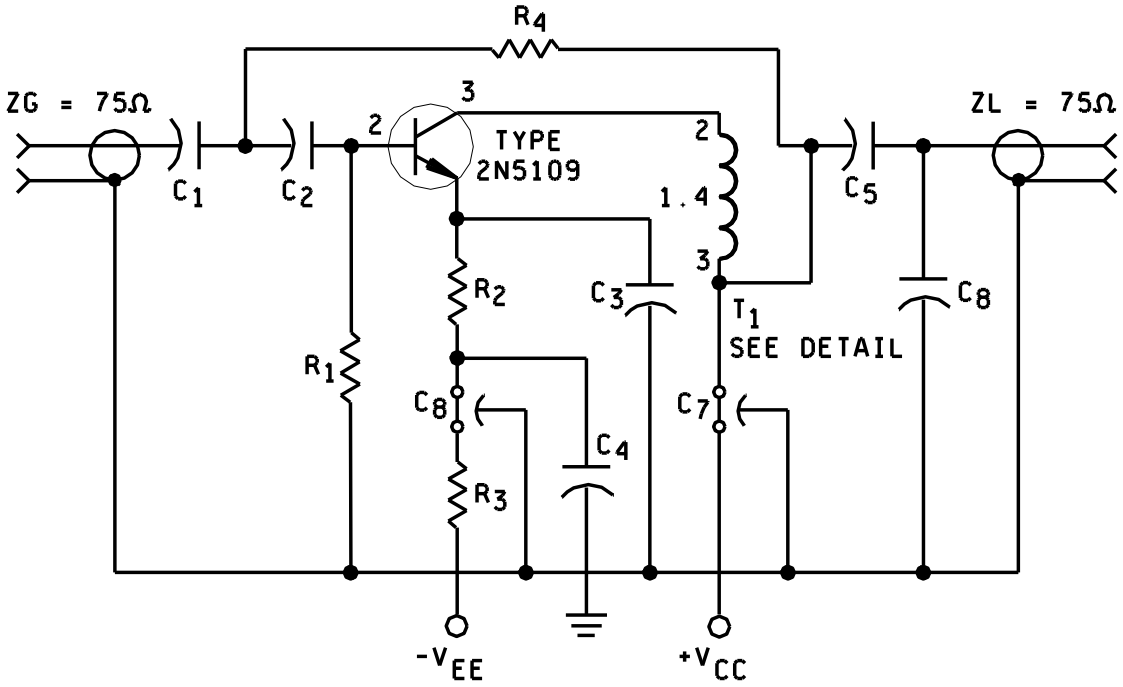


C_1, C_2, C_3: 1.0 to 30 pF.
C_4: 1.0 to 20 pF.
C_5: 10,000 pF.
C_6, C_7: 1,000 pF.
C_8: .01 µF.

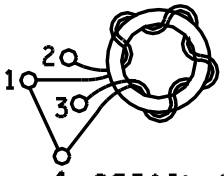
L_1: 4.5 turns, number 22 wire, .187 inch I. D.
L_4: 3.5 turns, number 22 wire, .187 inch I. D.
L_2, L_3: 0.82 µH RFC.
R_1: 240Ω, 2 watts.

FIGURE 3. RF amplifier for power gain and noise figure tests.

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C_1, C_2, C_3, C_5: 0.002 µF.
R_3 : 330Ω, 1 W.
C_4: 0.03 µF.
R_4 : 200Ω, 0.25 W.
C_6, C_7: 1500 pF.
T_1 : 4 turns number 30 wire.
C_8: 18 pF.
R_1: 4.7 kΩ, 0.25 W.
R_2: 6.8Ω, 0.5 W.



DETAIL OF TRANSFORMER T_1

Bifilar wound on "Indiana General".
Core number CF-102-Q1, or equivalent.

FIGURE 4. RF amplifier for voltage gain test.

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4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1).

c. Lead finish (see 3.3.1).

d. Product assurance level and type designator.

6.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue because of the extensiveness of the changes.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL-19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, DSCC-VQE, Columbus, OH 43216.

CONCLUDING MATERIAL

Custodians:
Air Force - 17

Review activities:
Air Force - 19, 85, 99

Preparing activity:
DLA - CC

(Project 5961-1934)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-19500/453C	**2. DOCUMENT DATE** 15 December 1997

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, HIGH-FREQUENCY TYPE 2N5109, JAN, JANTX, AND JANTXV

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) Commercial: DSN: FAX: EMAIL:	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. POINT OF CONTACT Alan Barone	b. TELEPHONE, FAX (Include Area Code), E-mail: (1) Commercial DSN FAX E-mail 614-692-0510 850-0510 614-692-6939 alan_barone@dscc.dla.mil
c. ADDRESS Defense Supply Center Columbus ATTN: DSCC-VAT Columbus, OH 43216-5000	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 DSN 289-2340

DD Form 1426, OCT 89 Previous editions are obsolete 198/290

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